

## 5.1 - MI

$P(n) \rightarrow \boxed{MI} \rightarrow \text{true for } n \in \mathbb{Z}$

$\rightarrow P(1)$  is true

$\rightarrow \forall k \in \mathbb{Z}^+ \quad P(k+1)$  is true iff  $P(k)$  is true

⑧  $\boxed{1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}} \rightarrow P(n)$

1.  $P(1)$  is true as L.H.S = 1, R.H.S =  $\frac{1(1+1)}{2} = 1$  — (i)

2.  $P(k) \rightarrow 1 + 2 + 3 + \dots + k = \frac{k(k+1)}{2} \rightarrow$  (Assume) true. — (ii)

3. to prove  $P(k)$  is true  $\rightarrow$  prove  $P(k+1)$  is also true

$$P(k+1) = 1 + 2 + 3 + \dots + (k+1) = \frac{(k+1)(k+2)}{2} \text{ — (iii)}$$

$$\begin{aligned} 1 + 2 + 3 + \dots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2} \end{aligned}$$

$P(n)$  is true for all  $n \in \mathbb{Z}$

Ex 2  $P(n) = \boxed{n < 2^n}$  all  $n \in \mathbb{Z}^+$

1.  $P(1) \rightarrow \boxed{1 < 2^1} \rightarrow \text{True.}$

2.  $P(k) \rightarrow k < 2^k \rightarrow \text{Assume true}$

3.  $P(k+1) \rightarrow (k+1) < 2^{k+1} =$

Adding 1 on both sides of (2)

$$P(k+1) \rightarrow \boxed{(k+1) < \underline{2^k + 1} < 2^k + 2^k}$$

$$= \begin{aligned} &\rightarrow (k+1) < \boxed{2^k + 2^k} \\ &\rightarrow (k+1) < \boxed{2 \cdot 2^k} \\ &\rightarrow (k+1) < 2^{k+1} \end{aligned}$$

$$2 < 3 \quad \text{and} \quad 3 < 5$$

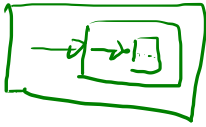
$$\rightarrow 2 < 3 < 5$$

$$\rightarrow (k+1) < 2^{k+1} < 2^k + 2^{k+1}$$

$$P(n) = n < 2^n$$

is true for all  $n \in \mathbb{Z}^+$

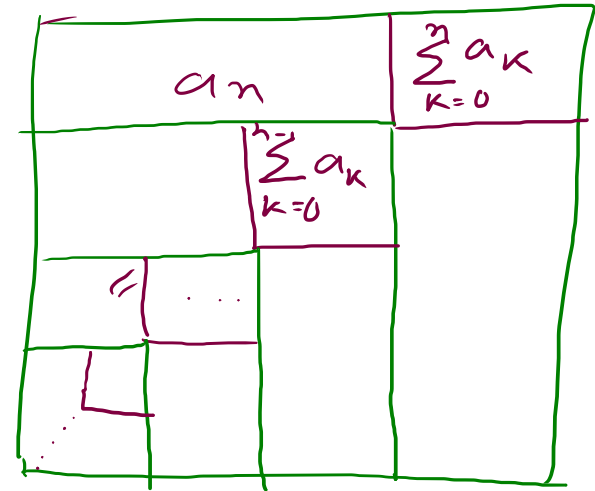
### 5.3 Recursive definitions.



Ex-1  $\sum_{k=0}^n a_k \rightarrow$  Give RD of this.

$$\rightarrow \sum_{k=0}^0 a_k = \underline{\underline{a_0}}$$

$$\sum_{k=0}^n a_k = a_n + \sum_{k=0}^{n-1} a_k$$



## 5.4 Recursive Algorithms.

② find factorial of  $n$  using RA

$$\begin{aligned} \hookrightarrow 1! &\rightarrow 1 & 5! &\rightarrow \boxed{1 \cdot 2 \cdot 3 \cdot 4} \cdot 5 \rightarrow 4! \cdot 5 \\ \underline{0!} &\rightarrow \textcircled{1} & 4! &\rightarrow \underline{1 \cdot 2 \cdot 3 \cdot 4} \end{aligned}$$

$$\begin{aligned} \text{fact}(n) &\rightarrow n \cdot \text{fact}(n-1) \\ &\quad \hookrightarrow (n-1) \cdot \text{fact}(\underline{n-2}) \dots 0 \end{aligned}$$

$\hookrightarrow$  procedure  $\text{fact}(n: n \in \mathbb{Z}^+)$   
if  $\boxed{n=0}$ . return 1  
else return  $n \cdot \text{fact}(n-1)$

①  $a^n \rightarrow RA$

$2^3 \rightarrow 8 \rightarrow 2 \cdot 2^2$

$\downarrow 2$

$2^1$

$\downarrow 2 \cdot 1 \equiv 2 \cdot 2^0$

$\text{power}(3, 3)$

$\downarrow 3 \cdot \text{power}(3, 2)$

$\downarrow 27$

$\downarrow 3 \cdot \text{power}(3, 1)$

$\downarrow 3$

$\downarrow 3 \cdot \text{power}(3, 0)$

$3^3 = 27$

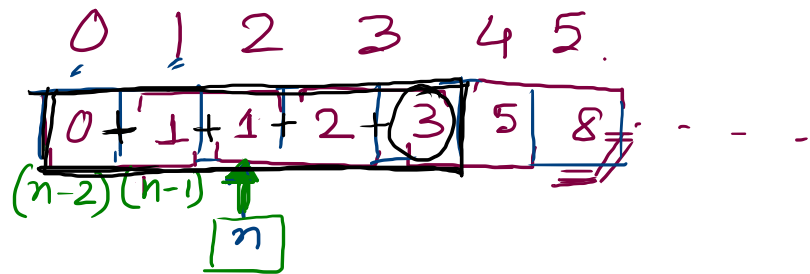
$3 \cdot 1$

$a^n \rightarrow a \cdot a^{n-1}$

↳ Procedure  $\text{power}(a, n)$

→ if  $n=0$  return 1  
else return  $a \cdot \text{power}(a, n-1)$

# ④ Fibonacci using RA.



↳ procedure  $\text{fib}(n: \mathbb{Z}^+)$   
 = if  $n=0 \rightarrow \text{return } 0$  -  
 = elseif  $n=1 \rightarrow \text{return } 1$  -  
 = else return  $\text{fib}(n-1) + \text{fib}(n-2)$

$\text{fib}(2)$   
 ③  $\text{fib}(1) + \text{fib}(0)$

Diagram showing the recursive call for  $\text{fib}(2)$ . The value 1 is shown below  $\text{fib}(1)$  and 0 is shown below  $\text{fib}(0)$ .



$\text{fib}(4) \rightarrow 2 + 1 = 3$   
 $\text{fib}(4-1) + \text{fib}(4-2)$

